EXPERIMENT 04

Finite-State Machine for Plural Noun Forms

Aim

To implement a simple finite-state-style rule system that generates plural forms of English nouns.

Procedure

1. Define rules for nouns ending in s, x, ch, or sh.
2. Add 'es' for these endings.
3. Change consonant plus y to 'ies'.
4. Add 's' for other nouns.
5. Test the rules using sample nouns.

Program

```
def plural(noun):
    lower = noun.lower()
    if lower.endswith(("s", "x", "ch", "sh")):
        return noun + "es"
    if lower.endswith("y") and len(noun) > 1 and lower[-2] not in "aeiou":
        return noun[:-1] + "ies"
    return noun + "s"

for noun in ["cat", "box", "church", "baby", "book"]:
    print(noun, "->", plural(noun))
```

Output

```
OUTPUT — EXPERIMENT 04
cat -> cats
box -> boxes
church -> churches
baby -> babies
book -> books
```

Result

Thus, a rule-based finite-state-style machine generated plural forms for the sample nouns.